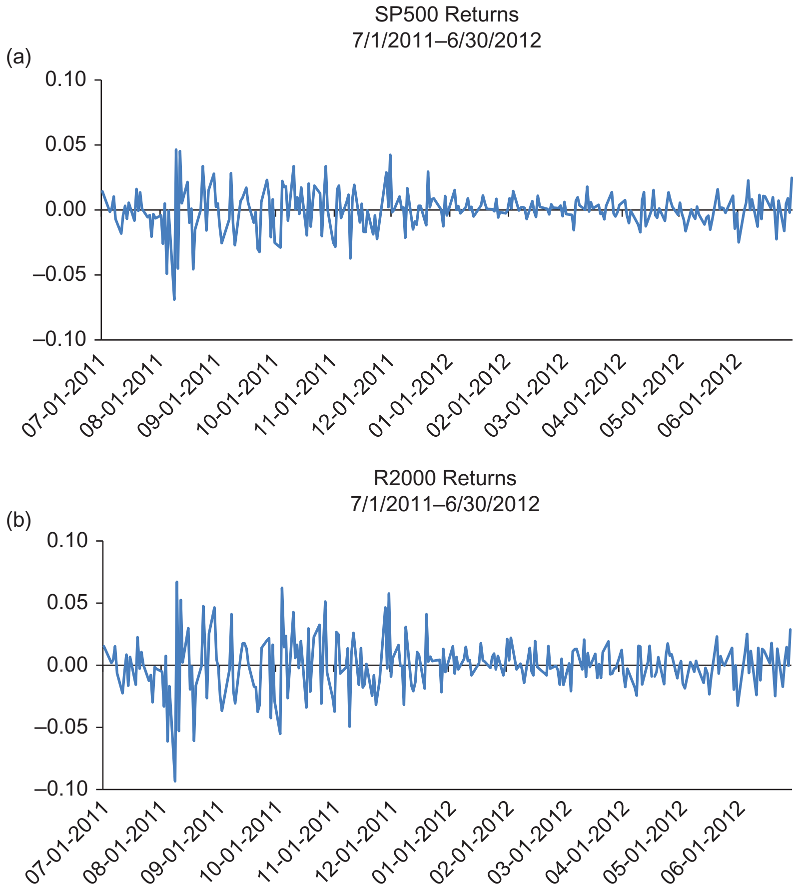

Figure 6.3 Daily Index Returns.

end. This was then followed by a relatively low volatility period from January 2012 through the end of May 2012 when markets started to experience increasing volatility levels. The most important finding over the time period investigated is that the volatility of both indexes appears to be time varying with clustering. The volatility model needs to be able to quickly adjust to these types of sudden regime shifts.

Historical Moving Average (HMA)

The historical moving average volatility measure is computed by definition:

$$\overline{\sigma}_t^2 = \frac{1}{n-1}\sum_{k=1}^{n} y_{t-k}^2 \tag{6.16}$$